Mathematical Programming Computational manuscript No.
(will be inserted by the editor)

Practical Regularized Quasi-Newton Methods with Inexact Function Values

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Received: date / Accepted: date

Abstract Many practical optimization problems involve objective function values that are corrupted by bounded, nonvanishing numerical errors. We propose a noise-tolerant regularized quasi-Newton method that exploits a relaxed Armijo line search when observed function values provide reliable progress and switches to an objective-function-free regularization mechanism otherwise. Under exact gradients, smooth nonconvexity, bounded relative-or-absolute function errors, and uniformly bounded quasi-Newton matrices, we prove an $\mathcal{O}\left(1/\varepsilon^2\right)$ iteration complexity for reaching first-order stationarity. The exact-gradient qualification is essential to the stated theorem; experiments with perturbed gradients are reported separately as deliberately adverse robustness tests beyond its scope. Experiments on CUTEst include function-value-only noise, joint function-and-gradient noise, and 64-, 32-, and 16-bit arithmetic. They evaluate both robustness and the sensitivity to the supplied function-error bound.

Keywords Regularized Quasi-Newton Method · Numerical Stability · Numerical Error · AdaGrad-Norm · Objective-Function-Free Optimization

Mathematics Subject Classification (2020) 90C53 · 90C06 · 65K05 · 49M15

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1 Introduction

We consider the unconstrained optimization problem

$$\underset{x\in\mathbb{R}^n}{\text{minimize}} \quad f(x), \tag{1}$$

where $f\colon \mathbb{R}^n \to \mathbb{R}$ is a continuously differentiable nonconvex function, and $x \in \mathbb{R}^n$ is the optimization variable. We focus on settings where objective function evaluations are contaminated by bounded and non-diminishing numerical noise [39]. Specifically, only inexact function values $\overline{f}(x)$ are available for optimization. Such inaccuracies are unavoidable in many realistic scenarios, including finite-precision arithmetic, simulation-based evaluations, and stochastic approximations [6, 29, 38]. When function values are contaminated by numerical noise, standard optimization algorithms may lose their theoretical guarantees, behave erratically, or terminate prematurely with unreliable solutions.

In the absence of noise and with accurate function evaluations, quasi-Newton methods, in particular the Broyden–Fletcher–Goldfarb–Shanno (BFGS) method and its Limited-memory variant (L-BFGS), are among the most efficient and widely used algorithms to solve the problem (1) [26, 31]. Their success is largely attributed to fast local convergence and scalability. Standard implementations [32, 42] rely on line search procedures that enforce the Wolfe conditions to guarantee sufficient descent and the positive definiteness of the Hessian approximation.

In the presence of numerical noise, these line search conditions may become unreliable. Differences in function values or directional derivatives can be dominated by numerical errors, leading to unstable step sizes, ill-conditioned Hessian approximations, or abnormal termination [33, 38]. From a computational perspective, improving the robustness of quasi-Newton methods in noisy environments is therefore crucial. In particular, it is desirable to design algorithms that (i) retain computational efficiency when function evaluations are accurate, (ii) remain stable when function values are unreliable, and (iii) guarantee convergence to first-order stationary points under mild assumptions.

In this work, we propose a noise-tolerant regularized quasi-Newton method that addresses these challenges. Our approach combines an Armijo-type line search with quadratic regularization [24, 41] and an Objective-Function-Free Optimization (OFFO) inspired strategy [18, 20]. Unlike the regularized L-BFGS method in [24], which establishes global convergence without an evaluation-complexity bound, our analysis gives the standard $\mathcal{O}\left(1/\varepsilon^2\right)$ complexity under bounded function-value errors. Unlike OFFO algorithms, which deliberately never evaluate the objective function, our hybrid method retains a relaxed line search whenever observed function values certify progress and invokes OFFO-type scaling only otherwise. The noise-tolerant quasi-Newton analysis in [38] yields a stationarity neighborhood whose function-error contribution is proportional to $\sqrt{M\varepsilon_f}$, whereas its gradient-error contribution is proportional to ε_g. Consequently, up to problem-dependent constants, making the former no larger than the latter requires the comparatively restrictive regime $\varepsilon_f = \mathcal{O}\left(\varepsilon_g^2/M\right)$. Our focus is complementary: function values may carry substantial nonvanishing error, as is common in low-precision computation, while gradients are exact in the complexity theorem.

When a sufficient decrease in observed function value is available, the method uses an unregularized quasi-Newton step; otherwise, it uses regularization governed by a phasewise AdaGrad-Norm accumulator. We prove that the resulting method reaches an ε-first-order stationary point in $\mathcal{O}\left(1/\varepsilon^2\right)$ iterations for smooth nonconvex objectives under exact gradients and bounded relative-or-absolute function errors. We state the exact-gradient scope explicitly and separate theory-aligned function-noise experiments from joint function-and-gradient perturbations, the latter serving as deliberately adverse stress tests rather than evidence for the theorem. The CUTEst study also reports problem dimensions, the baseline stopping rules, sensitivity to the supplied function-error estimate, and ablations of the hybrid mechanism.

The remainder of this paper is organized as follows. In section 2, we review relevant background and introduce the problem setting. In section 3, we present the proposed algorithm. In section 4, we establish a global convergence property of the proposed algorithm. In section 5, we discuss practical choices of algorithmic variables and parameters. In section 6, we report numerical results on the CUTEst benchmark collection. Finally, in section 7, we conclude the paper with a summary and future research directions.

Our implementation is publicly available on GitHub[1].

2 Preliminaries and Problem Setting

We begin by reviewing relevant background and formalizing the problem setting.

2.1 Regularized Quasi-Newton Method

The regularized quasi-Newton method is a variant of quasi-Newton methods that does not rely heavily on line search techniques. This property is important for optimization in noisy evaluation environments. There has been renewed interest in regularized Newton-type methods, including cubic-regularization schemes [30, 35, 43], and we focus on quadratic regularization [10, 24, 25, 37, 40, 41, 48] for simplicity of analysis and implementation.

For $g_k := \nabla f(x_k)$ and an approximate Hessian $B_k \in \mathbb{R}^{n\times n}$, regularized quasi-Newton methods compute the step as

$$x_{k+1} = x_k + d_k, \quad d_k = -(B_k + \mu_k I)^{-1} g_k,$$

with a regularization parameter $\mu_k \geq 0$. Combining regularization with line search is also possible and has been explored [40]. Several strategies for choosing μ_k in nonconvex optimization has been proposed [37, 40, 41], and one of them is given by

$$\mu_k = c_1 \max(0, -\lambda_{\min}(B_k)) + c_2 \|\nabla f(x_k)\|^{\delta},$$

where $c_1 > 1$, $c_2 > 0$, $\delta \geq 0$, and $\lambda_{\min}(B_k)$ denotes the smallest eigenvalue of B_k. This choice guarantees that $B_k + \mu_k I$ is positive definite, ensuring that d_k is a descent direction. We adapt such regularization strategies to noisy evaluation environments.

[1] https://github.com/HirokiHamaguchi/qnlab

2.2 Objective-Function-Free Optimization

Objective-Function-Free Optimization (OFFO) refers to a class of methods that avoid explicit evaluations of the objective function and rely solely on gradient information [18, 20]. Such methods are particularly attractive when function values are unreliable due to noise. Similar adaptive trust-region approaches are also discussed [17].

Prominent examples of OFFO include adaptive gradient methods such as AdaGrad [12] and AdaGrad-Norm [44]. Their update rules are

$$\text{(AdaGrad)} \quad x_{k+1}^{(i)} = x_k^{(i)} - \frac{1}{\theta\sqrt{\varsigma + \sum_{j=0}^{k}(g_j^{(i)})^2}} g_k^{(i)}, \qquad (i = 1,2,\ldots,n)$$

$$\text{(AdaGrad-Norm)} \quad x_{k+1} = x_k - \frac{1}{\theta\sqrt{\varsigma + \sum_{j=0}^{k}\|g_j\|^2}} g_k$$

respectively, where $\varsigma > 0$ and $\theta > 0$ are algorithmic parameters, and $x^{(i)}$ denotes the i-th component of a vector x.

2.3 Problem Setting

We consider optimization problems in which exact objective function values are unavailable, following the recent literature on error-aware optimization [2, 7, 8, 19, 36, 39]. When both objective values and gradients are heavily corrupted by noise, reliable optimization cannot be guaranteed. Thus, meaningful optimization requires that at least one source of information be sufficiently accurate. This work focuses on problems arising from finite-precision computations or inexact evaluations, where objective function evaluations are subject to non-negligible errors while gradient information can be computed with relatively high accuracy. Settings in which objective values are accurate but gradients are noisy, such as derivative-free optimization or stochastic approximation methods [3, 38, 45], are beyond the scope of this work.

In the following, we formalize the assumptions on function value and gradient evaluations. We assume that only an error-contaminated function value $\overline{f}(x)$ is accessible, satisfying the hybrid absolute-relative error model [22, 23]:

$$\left|f(x) - \overline{f}(x)\right| \leq \varepsilon_{\mathrm{f}} \max(1, |f(x)|), \tag{2}$$

where $0 \leq \varepsilon_{\mathrm{f}} < 1$ is an error rate. The error is predominantly relative when $|f(x)|$ is large and effectively absolute when $|f(x)|$ is small, capturing the typical behavior of rounding errors in IEEE 754 floating-point arithmetic [22]. In the numerical experiments, ε_{f} is taken as a large multiple of the machine epsilon to account for accumulated round-off effects. Closely related models are also employed in practical stopping criteria [42].

We distinguish the setting covered by the analysis from the additional stress tests in section 6. The convergence theory makes the following exact-gradient assumption.

Assumption 1 (Exact gradient) *For every evaluated point x,*

$$g(x) = \nabla \overline{f}(x) = \nabla f(x). \tag{3}$$

This assumption is not implied by the function-value error model in eq. (2) and is therefore stated independently. With a bounded nonvanishing gradient error, exact first-order stationarity is generally impossible to certify below the noise floor; a result in that setting would instead have to bound convergence to a neighborhood. Accordingly, function-value-only noise is the theory-aligned artificial experiment, while joint function-and-gradient noise is reported as a deliberately adverse robustness test beyond the theorem.

3 Proposed Algorithm

We propose a new algorithm for the problem setting stated in section 2.3. We draw inspiration from the OFFO framework to design robust regularization and scaling strategies. Unlike purely OFFO methods, our algorithm exploits function values when they are numerically reliable and smoothly switches to gradient-based control otherwise. This hybrid behavior enables improved stability without sacrificing efficiency in practical optimization tasks. The pseudo-code of our algorithm is given in Algorithm 1. The minimum of the empty set is defined as $+\infty$ in line 3. Algorithm 2 is an auxiliary line search algorithm called in Algorithm 1. In the following, we explain the three main components of Algorithm 1.

Algorithm 1: Noise Tolerant Regularized Quasi-Newton Method

Input: x_0, $0 < k_{\max}$, $0 \le \varepsilon_f < 1$, $0 < \theta_{\min} \le \theta_{\max}$ and $0 < \varsigma < 1$

1 $k \leftarrow 0$, $g_0 \leftarrow \nabla \overline{f}(x_0)$, $K^0 \leftarrow \emptyset$, $K^+ \leftarrow \emptyset$
2 **for** $k = 0, 1, 2, \ldots, k_{\max} - 1$ **do**
3 **if** $\min_{j \in K^0,\, j<k} (\overline{f}(x_j) - \Delta_j) \ge \overline{f}(x_k)$ **then**
4 $K^0 \leftarrow K^0 \cup \{k\}$
5 $\mu_k \leftarrow 0$
6 **else**
7 $K^+ \leftarrow K^+ \cup \{k\}$
8 $\mu_k \leftarrow \theta_k \sqrt{\varsigma + \sum_{j \in K^+,\, j \le k} \|g_j\|^2}$ with $\theta_k \in [\theta_{\min}, \theta_{\max}]$ (see section 5.3).
9 $d_k \leftarrow -(B_k + \mu_k I)^{-1} g_k$ with B_k (see section 5.1).
10 Find α_k and Δ_k satisfying eq. (4) by Algorithm 2.
11 $x_{k+1} \leftarrow x_k + \alpha_k d_k$, $g_{k+1} \leftarrow \nabla \overline{f}(x_{k+1})$
12 **end**

Algorithm 2: Backtracking Line Search with Relaxed Armijo Condition

Input: $x_k \in \mathbb{R}^n$, $d_k \in \mathbb{R}^n$, $g_k \in \mathbb{R}^n$, $0 < c < 1$ and $0 < \beta_{\min} < \beta_{\max} < 1$

1 $\alpha_k \leftarrow 1$
2 $\Delta_k \leftarrow \frac{2\varepsilon_f}{1-\varepsilon_f} \max(1, \overline{f}(x_k), -\overline{f}(x_k + \alpha_k d_k))$
3 **while** $\overline{f}(x_k) + c\alpha_k g_k^\top d_k + \Delta_k < \overline{f}(x_k + \alpha_k d_k)$ **do**
4 $\quad \alpha_k \leftarrow \beta \alpha_k$ with $\beta \in [\beta_{\min}, \beta_{\max}]$ (see section 5.2).
5 $\quad \Delta_k \leftarrow \frac{2\varepsilon_f}{1-\varepsilon_f} \max(1, \overline{f}(x_k), -\overline{f}(x_k + \alpha_k d_k))$
6 **end**
7 **return** α_k

Computation of the Regularized Quasi-Newton Direction (line 9): The first component is the computation of the regularized quasi-Newton direction d_k stated in section 2.1. Given $g_k = \nabla \overline{f}(x_k) = \nabla f(x_k)$, an approximate Hessian B_k, and a regularization parameter μ_k, we compute the search direction $d_k = -(B_k + \mu_k I)^{-1} g_k$. A practical choice of B_k is discussed in section 5.1.

Relaxed Armijo Condition with an Error-Absorbing Term (line 10): The second component is the computation of the step size α_k using a relaxed Armijo condition, which is defined as follows:

$$\begin{cases} \overline{f}(x_k) + c\alpha_k g_k^\top d_k + \Delta_k \geq \overline{f}(x_k + \alpha_k d_k), \\ \Delta_k = \dfrac{2\varepsilon_f}{1-\varepsilon_f} \max(1, \overline{f}(x_k), -\overline{f}(x_k + \alpha_k d_k)), \end{cases} \tag{4}$$

where Δ_k is the error-absorbing term that is recomputed for each α_k. This relaxation guarantees the existence of $\alpha_k \in (0, 1]$ even in the presence of errors in $\overline{f}$. When d_k is a descent direction ($g_k^\top d_k < 0$), eq. (4) ensures that a temporary increase of $\overline{f}$ is at most Δ_k. Since Δ_k depends on $-\overline{f}(x_k + \alpha_k d_k)$ but not on $\overline{f}(x_k + \alpha_k d_k)$, Δ_k does not increase when $\overline{f}(x_k + \alpha_k d_k)$ becomes large. This prevents unbounded growth of Δ_k.

Adaptive Update of the Regularization Parameter (lines 3–8): The third component is the adaptive update of the regularization parameter μ_k. A larger μ_k yields a more conservative step, while a smaller μ_k allows a more aggressive quasi-Newton step, especially when $\mu_k = 0$. In practical settings, $\mu_k = 0$ is preferred, since this allows us to fully exploit the quasi-Newton approximation and typically leads to practically faster convergence. Let us partition the iteration indices $K := \{0, 1, 2, \ldots, k_{\max} - 1\}$ into two sets:

$$K^0 := \{k \in K \mid \mu_k = 0\}, \quad K^+ := \{k \in K \mid \mu_k > 0\}. \tag{5}$$

We set $\mu_k = 0$ if and only if the following condition is satisfied:

$$\min_{j \in K^0,\, j < k} \left(\overline{f}(x_j) - \Delta_j\right) \geq \overline{f}(x_k). \tag{6}$$

Note that $k = 0$ satisfies eq. (6) since K^0 is empty. This condition restricts $\mu_k = 0$ to iterations for which a sufficient decrease in $\overline{f}$ has been observed. Thereby, while $\overline{f}$ can temporarily increase, this condition theoretically ensures that $\overline{f}$ does not grow unboundedly or oscillate indefinitely. Otherwise, to ensure convergence to first-order stationary points, we apply an OFFO-based update

$$\mu_k = \theta_k \sqrt{\varsigma + \sum_{j \in K^+, j \le k} \|g_j\|^2}, \tag{7}$$

as described in section 2.2. Since OFFO schemes do not rely on function values, this update is robust to numerical errors in $\overline{f}$.

4 Theoretical Analysis

In this section, we establish the global convergence properties of Algorithm 1 under the problem setting in section 2.3, namely, eqs. (2) and (3).

4.1 Assumptions

In addition to the exact-gradient assumption stated in Assumption 1, we make the following three structural assumptions. The first assumption ensures that the objective function is bounded below.

Assumption 2 *The function f is bounded below, meaning that for all $x \in \mathbb{R}^n$,*

$$f(x) \ge f^* := \inf_{x \in \mathbb{R}^n} f(x) > -\infty.$$

By the triangle inequality and eq. (2), we have $\overline{f}(x) \ge f(x) - \varepsilon_f \max(1, |f(x)|)$. Since the function $g(t) = t - \varepsilon_f \max(1, |t|)$ is non-decreasing for $0 \le \varepsilon_f < 1$, it follows that $\inf_x g(f(x)) = g(f^*)$. Consequently, we can also lower bound $\overline{f}$ as

$$\overline{f}(x) \ge \overline{f}^* := \inf_{x \in \mathbb{R}^n} (f(x) - \varepsilon_f \max(1, |f(x)|)) = f^* - \varepsilon_f \max(1, |f^*|) > -\infty. \tag{8}$$

The second assumption ensures smoothness of the objective function.

Assumption 3 *The function $f\colon \mathbb{R}^n \to \mathbb{R}$ is L-smooth, meaning that there exists a constant $L > 0$ such that for all $x, y \in \mathbb{R}^n$,*

$$f(y) \le f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2} \|y - x\|^2.$$

The third assumption requires the matrices $\{B_k\}_{k=0}^{k_{\max}-1}$ to be uniformly positive definite and bounded.

Assumption 4 *There exist constants $0 < m \le M$ such that all B_k satisfy*

$$mI \preceq B_k \preceq MI.$$

As we will discuss in section 5, the standard BFGS update with a slight modification guarantees Assumption 4 in practice. By Assumption 4, we can derive useful bounds involving g_k and d_k. Using $d_k = -(B_k+\mu_k I)^{-1} g_k$ (line 9) and Assumption 4 asserting $(m+\mu_k)I \preceq B_k + \mu_k I \preceq (M+\mu_k)I$, we obtain the following bounds:

$$-g_k^\top d_k = d_k^\top (B_k+\mu_k I) d_k \geq (m+\mu_k)\|d_k\|^2 \geq m\|d_k\|^2, \tag{9}$$

$$-g_k^\top d_k = g_k^\top (B_k+\mu_k I)^{-1} g_k \geq \frac{1}{M+\mu_k}\|g_k\|^2, \tag{10}$$

$$\|d_k\|^2 = \left\|(B_k+\mu_k I)^{-1} g_k\right\|^2 \leq \frac{1}{(m+\mu_k)^2}\|g_k\|^2. \tag{11}$$

4.2 Error Bound on Function Evaluations

We first establish useful properties of the error-absorbing term Δ_k defined in eq. (4). In the following, we define the actual and observed function decreases as

$$\delta_k := f(x_k) - f(x_{k+1}), \tag{12}$$

$$\overline{\delta}_k := \overline{f}(x_k) - \overline{f}(x_{k+1}). \tag{13}$$

The first property states that the error between the actual values difference δ_k and the observed values difference $\overline{\delta}_k$ can be bounded by Δ_k.

Lemma 1 *If x_k and x_{k+1} satisfy $f(x_{k+1}) \leq f(x_k)$, it holds that*

$$\delta_k \leq \overline{\delta}_k + \Delta_k.$$

Proof We start from general observations. We first prove that, for all $x \in \mathbb{R}^n$,

$$\begin{cases} \max(1, +f(x)) \leq \frac{1}{1-\varepsilon_f} \max(1, +\overline{f}(x)), \\ \max(1, -f(x)) \leq \frac{1}{1-\varepsilon_f} \max(1, -\overline{f}(x)). \end{cases} \tag{14}$$

We prove the inequality with the plus sign. If $+f(x) \leq 1$, then $\max(1,+f(x)) = 1 \leq \frac{1}{1-\varepsilon_f} \leq \frac{1}{1-\varepsilon_f}\max(1,+\overline{f}(x))$. If $+f(x) > 1$, eq. (2) implies

$$\left|f(x) - \overline{f}(x)\right| \leq \varepsilon_f \max(1, |f(x)|) = +\varepsilon_f f(x) \tag{15}$$

and thus $f(x) - \overline{f}(x) \leq +\varepsilon_f f(x)$. Hence,

$$\max(1,+f(x)) = +f(x) \leq +\frac{1}{1-\varepsilon_f}\overline{f}(x) \leq \frac{1}{1-\varepsilon_f}\max(1,+\overline{f}(x)).$$

The case with the negative sign can be shown analogously, using $-f(x) + \overline{f}(x) \leq -\varepsilon_f f(x)$, which follows from eq. (15). This completes the proof of eq. (14).

We also have the following general bound for all $k \in K$:

$$\begin{aligned} \left|\delta_k - \overline{\delta}_k\right| &= \left|(f(x_k) - \overline{f}(x_k)) - (f(x_{k+1}) - \overline{f}(x_{k+1}))\right| && \text{(rearrangement)} \\ &\leq \left|f(x_k) - \overline{f}(x_k)\right| + \left|f(x_{k+1}) - \overline{f}(x_{k+1})\right| && \text{(triangle inequality)} \end{aligned}$$

$$\begin{aligned}
&\leq \varepsilon_{\mathrm{f}} \max(1, |f(x_k)|) + \varepsilon_{\mathrm{f}} \max(1, |f(x_{k+1})|) && \text{(eq. (2))} \\
&\leq 2\varepsilon_{\mathrm{f}} \max(1, |f(x_k)|, |f(x_{k+1})|). && \text{(max property)} \qquad (16)
\end{aligned}$$

With these preparations, we now prove the lemma. From the condition $f(x_{k+1}) \leq f(x_k)$, eq. (16) can be further bounded as

$$\begin{aligned}
2\varepsilon_{\mathrm{f}} \max(1, |f(x_k)|, |f(x_{k+1})|) &= 2\varepsilon_{\mathrm{f}} \max(1, f(x_k), -f(x_{k+1})) && (f(x_{k+1}) \leq f(x_k)) \\
&\leq \frac{2\varepsilon_{\mathrm{f}}}{1-\varepsilon_{\mathrm{f}}} \max(1, \overline{f}(x_k), -\overline{f}(x_{k+1})) && \text{(eq. (14))} \\
&= \Delta_k, && \text{(eq. (4))}
\end{aligned}$$

which yields the desired inequality with $\delta_k - \overline{\delta}_k \leq \left| \delta_k - \overline{\delta}_k \right| \leq \Delta_k$. □

We also derive the following lemma for later use.

Lemma 2 *Under Assumption 4, it holds that*

$$\overline{\delta}_k \leq \delta_k + \frac{1+\varepsilon_{\mathrm{f}}}{1-\varepsilon_{\mathrm{f}}} \Delta_k.$$

Proof Since $-g_k^\top d_k \geq 0$ from Assumption 4 and eq. (9), the relaxed Armijo condition (eq. (4)) implies

$$\overline{f}(x_k) - \overline{f}(x_{k+1}) \geq -c\alpha_k g_k^\top d_k - \Delta_k \geq -\Delta_k. \qquad (17)$$

Thus, using the general results in eq. (16) and $\Delta_k \geq 0$, we can further bound as

$$\begin{aligned}
\left| \delta_k - \overline{\delta}_k \right| &\leq 2\varepsilon_{\mathrm{f}} \max(1, |f(x_k)|, |f(x_{k+1})|) && \text{(eq. (16))} \\
&\leq \frac{2\varepsilon_{\mathrm{f}}}{1-\varepsilon_{\mathrm{f}}} \max(1, \overline{f}(x_k), -\overline{f}(x_k), \overline{f}(x_{k+1}), -\overline{f}(x_{k+1})) && \text{(eq. (14))} \\
&\leq \frac{2\varepsilon_{\mathrm{f}}}{1-\varepsilon_{\mathrm{f}}} \max(1, \overline{f}(x_k) + \Delta_k, -\overline{f}(x_{k+1}) + \Delta_k) && \text{(eq. (17))} \\
&\leq \frac{2\varepsilon_{\mathrm{f}}}{1-\varepsilon_{\mathrm{f}}} \left(\max(1, \overline{f}(x_k), -\overline{f}(x_{k+1})) + \Delta_k \right) \\
&= \left(1 + \frac{2\varepsilon_{\mathrm{f}}}{1-\varepsilon_{\mathrm{f}}} \right) \Delta_k = \frac{1+\varepsilon_{\mathrm{f}}}{1-\varepsilon_{\mathrm{f}}} \Delta_k, && \text{(eq. (4))}
\end{aligned}$$

which yields the desired inequality with $\overline{\delta}_k - \delta_k \leq \left| \delta_k - \overline{\delta}_k \right| \leq \frac{1+\varepsilon_{\mathrm{f}}}{1-\varepsilon_{\mathrm{f}}} \Delta_k$. □

4.3 Backtracking Stage

We next analyze the line search in Algorithm 2. Recall that the parameter c in Algorithm 2 satisfies $0 < c < 1$ and $0 < m$ from Assumption 4.

Lemma 3 *Under Assumptions 3 and 4, the backtracking procedure in Algorithm 2 terminates whenever the step size α_k satisfies*

$$\alpha_k \leq \frac{2(1-c)m}{L}. \tag{18}$$

Proof Under the L-smoothness of f (Assumption 3), we always have

$$\begin{aligned} f(x_k) - f(x_k + \alpha_k d_k) &\geq -\alpha_k g_k^\top d_k - \frac{L}{2}\alpha_k^2 \|d_k\|^2 \\ &= -c\alpha_k g_k^\top d_k + \alpha_k \left(-(1-c) g_k^\top d_k - \frac{L\alpha_k}{2}\|d_k\|^2 \right). \end{aligned} \tag{19}$$

From Assumption 4 and eq. (9), we have $-g_k^\top d_k \geq m\|d_k\|^2$. Under eq. (18), we have

$$-(1-c) g_k^\top d_k - \frac{L\alpha_k}{2}\|d_k\|^2 \geq \left((1-c)m - \frac{L\alpha_k}{2} \right) \|d_k\|^2 \geq 0. \tag{20}$$

Combining eqs. (19) and (20) yields

$$f(x_k) - f(x_k + \alpha_k d_k) \geq -c\alpha_k g_k^\top d_k.$$

Since $-g_k^\top d_k \geq 0$ from Assumption 4 and eq. (9), the condition of Lemma 1 is satisfied with $x_{k+1} = x_k + \alpha_k d_k$. Thus we obtain $\delta_k \leq \overline{\delta}_k + \Delta_k$, i.e.,

$$-c\alpha_k g_k^\top d_k \leq f(x_k) - f(x_k + \alpha_k d_k) \leq \overline{f}(x_k) - \overline{f}(x_k + \alpha_k d_k) + \Delta_k.$$

This inequality means that the relaxed Armijo condition (eq. (4)) is satisfied and thus the backtracking procedure terminates. This completes the proof. □

Now, let us analyze Algorithm 2 using Lemma 3. Define

$$\overline{\alpha} := \min\left(\beta_{\min} \frac{2(1-c)m}{L}, 1 \right) > 0.$$

The following states the key property of the backtracking procedure.

Proposition 1 *Under Assumptions 3 and 4, the backtracking line search in Algorithm 2 terminates after finitely many rejections, and the step size α_k satisfies*

$$\alpha_k \geq \overline{\alpha}.$$

Proof The line 4 in Algorithm 2 multiplies α_k by β at each rejection, satisfying $0 < \beta_{\min} \leq \beta \leq \beta_{\max} < 1$. For the number of backtracking rejections r, we have $\alpha_k \leq \beta_{\max}^r$. Thus, r is finite since the backtracking procedure terminates for sufficiently small α_k by Lemma 3. If $r = 0$, $\alpha_k = 1 \geq \overline{\alpha}$. Otherwise, $\alpha_k \geq \beta_{\min}\alpha_k'$ where α_k' is the last rejected step size. By Lemma 3, we have $\alpha_k' > \frac{2(1-c)m}{L}$, and thus

$$\alpha_k \geq \beta_{\min}\alpha_k' > \beta_{\min}\frac{2(1-c)m}{L} \geq \overline{\alpha}.$$

This completes the proof. □

4.4 Bound for the case $\mu_k = 0$

In this subsection, we lower bound the sum of actual function decreases δ_k over $k \in K^0$. Recall that K^0 denotes the set of iteration indices k such that $\mu_k = 0$, as in eq. (5). To this end, we establish that the sum of Δ_k over $k \in K^0$ is upper bounded, where Δ_k is the error-absorbing term in the relaxed Armijo condition (eq. (4)). We introduce the following constant $\Delta_{\max}$, which serves as an upper bound for Δ_k of $k \in K^0$.

$$\Delta_{\max} := \frac{2\varepsilon_f}{1-\varepsilon_f} \max(1, \overline{f}(x_0), -\overline{f}^*).$$

Lemma 4 *Under Assumptions 2 to 4, the sum of Δ_k over $k \in K^0$ satisfies*

$$\sum_{k \in K^0} \Delta_k \leq \overline{f}(x_0) - \overline{f}^* + \Delta_{\max}.$$

Proof Let $(k_i)_{i=0}^{\ell}$ with $\ell \geq 0$ be the elements of K^0 in increasing order. We have $\overline{f}(x_{k_\ell}) \leq \overline{f}(x_0)$ from eq. (6) with $k = k_\ell$, where the case $\ell = 0$ follows from $k_0 = 0$. Combining this with Assumption 2, we have $\Delta_{k_\ell} \leq \Delta_{\max}$. For all $i < \ell$, eq. (6) with $j = k_i$ and $k = k_{i+1}$ yields

$$\Delta_{k_i} \leq \overline{f}(x_{k_i}) - \overline{f}(x_{k_{i+1}}), \tag{21}$$

and summing Δ_k over $k \in K^0$ gives

$$\begin{aligned}
\sum_{k \in K^0} \Delta_k &\leq \sum_{i=0}^{\ell-1} (\overline{f}(x_{k_i}) - \overline{f}(x_{k_{i+1}})) + \Delta_{k_\ell} && \text{(eq. (21))} \\
&= \overline{f}(x_0) - \overline{f}(x_{k_\ell}) + \Delta_{k_\ell} && \text{(telescoping sum)} \\
&\leq \overline{f}(x_0) - \overline{f}^* + \Delta_{\max}. && (-\overline{f}(x_{k_\ell}) \leq -\overline{f}^* \text{ and } \Delta_{k_\ell} \leq \Delta_{\max})
\end{aligned}$$

This completes the proof. □

We now bound the sum of $\delta_k = f(x_k) - f(x_{k+1})$ over $k \in K^0$ with $\|g_k\|^2$ as follows.

Lemma 5 *Under Assumptions 2 to 4, the sum of δ_k over $k \in K^0$ satisfies*

$$\sum_{k \in K^0} \delta_k \geq \frac{c\overline{\alpha}}{M} \sum_{k \in K^0} \|g_k\|^2 - \frac{2}{1-\varepsilon_f} \left(\overline{f}(x_0) - \overline{f}^* + \Delta_{\max} \right).$$

Proof The relaxed Armijo condition (eq. (4)) at iteration $k \in K^0$ yields

$$\begin{aligned}
\overline{\delta}_k + \Delta_k &= \overline{f}(x_k) - \overline{f}(x_{k+1}) + \Delta_k && \text{(eq. (13))} \\
&\geq -c\alpha_k g_k^\top d_k && \text{(eq. (4))} \\
&\geq \frac{c\overline{\alpha}}{M + \mu_k} \|g_k\|^2 && \text{(Proposition 1 and eq. (10))} \\
&= \frac{c\overline{\alpha}}{M} \|g_k\|^2. && (\mu_k = 0 \text{ since } k \in K^0)
\end{aligned} \tag{22}$$

Using Lemma 2, we also have

$$\overline{\delta}_k + \Delta_k \leq \delta_k + \frac{1+\varepsilon_f}{1-\varepsilon_f}\Delta_k + \Delta_k = \delta_k + \frac{2}{1-\varepsilon_f}\Delta_k. \tag{23}$$

Then, combining eqs. (22) and (23) and summing over $k \in K^0$ gives

$$\sum_{k\in K^0} \frac{c\overline{\alpha}}{M}\|g_k\|^2 \leq \sum_{k\in K^0} \delta_k + \frac{2}{1-\varepsilon_f}\sum_{k\in K^0}\Delta_k \leq \sum_{k\in K^0}\delta_k + \frac{2}{1-\varepsilon_f}\left(\overline{f}(x_0) - \overline{f}^* + \Delta_{\max}\right),$$

where the last inequality follows from Lemma 4. Rearranging this completes the proof. □

4.5 Bound for the case $\mu_k > 0$

Next, we show that the sum of δ_k over $k \in K^+$ is lower bounded. Recall that K^+ is the set of iteration indices k such that $\mu_k > 0$ as defined in eq. (5), and that μ_k is defined as $\theta_k\sqrt{\varsigma + \sum_{j\in K^+,\ j\leq k}\|g_j\|^2}$ with $\theta_k \in [\theta_{\min}, \theta_{\max}]$ in line 8 of Algorithm 1.

Before proceeding to the main proof, we present Lemma 6, providing standard inequalities in the AdaGrad-Norm algorithm analysis [44, Lemma 3.2], [18, Lemma 3.1]. Its proof is deferred to appendix A.

Lemma 6 *In Algorithm 1, we have*

$$\sum_{k\in K^+}\frac{\|g_k\|^2}{\mu_k^2} \leq \frac{1}{\theta_{\min}^2}\left(\ln\left(\varsigma + \sum_{k\in K^+}\|g_k\|^2\right) - \ln(\varsigma)\right),$$
$$\sum_{k\in K^+}\frac{\|g_k\|^2}{\mu_k} \geq \frac{1}{\theta_{\max}}\left(\sqrt{\varsigma + \sum_{k\in K^+}\|g_k\|^2} - \sqrt{\varsigma}\right).$$

We now lower bound the sum of δ_k over $k \in K^+$ with $\|g_k\|^2$. We define the following constants for later use:

$$C_1 := \frac{\overline{\alpha}}{\theta_{\max}}, \qquad C_2 := \frac{M\overline{\alpha} + L/2}{\theta_{\min}^2}.$$

Lemma 7 *Under Assumptions 3 and 4, the sum of δ_k over $k \in K^+$ satisfies*

$$\sum_{k\in K^+}\delta_k \geq C_1\left(\sqrt{\varsigma + \sum_{k\in K^+}\|g_k\|^2} - \sqrt{\varsigma}\right) - C_2\left(\ln\left(\varsigma + \sum_{k\in K^+}\|g_k\|^2\right) - \ln(\varsigma)\right).$$

Proof From the L-smoothness (Assumption 3), for $k \in K^+$ we have

$$\begin{aligned}
\delta_k &= f(x_k) - f(x_k + \alpha_k d_k) && \text{(eq. (12) and } x_{k+1} = x_k + \alpha_k d_k\text{)} \\
&\geq -\alpha_k g_k^\top d_k - \frac{\alpha_k^2 L}{2}\|d_k\|^2 && \text{(Assumption 3)}
\end{aligned}$$

$$\geq \left(\frac{\alpha_k}{M+\mu_k} - \frac{\alpha_k^2 L}{2(m+\mu_k)^2}\right)\|g_k\|^2 \qquad \text{(eqs. (10) and (11))}$$
$$\geq \left(\frac{\overline{\alpha}}{M+\mu_k} - \frac{L}{2(m+\mu_k)^2}\right)\|g_k\|^2. \qquad \text{(Proposition 1 and } \alpha_k \leq 1) \tag{24}$$

We can further bound the terms in eq. (24) as

$$\frac{\overline{\alpha}}{M+\mu_k} = \frac{\overline{\alpha}}{\mu_k}\frac{1}{1+\frac{M}{\mu_k}} \geq \frac{\overline{\alpha}}{\mu_k}\left(1-\frac{M}{\mu_k}\right) = \frac{\overline{\alpha}}{\mu_k} - \frac{M\overline{\alpha}}{\mu_k^2}, \quad \frac{L}{2(m+\mu_k)^2} \leq \frac{L}{2\mu_k^2}. \tag{25}$$

Thus, applying eq. (25) to eq. (24) yields

$$\delta_k \geq \left(\frac{\overline{\alpha}}{\mu_k} - \frac{M\overline{\alpha}+L/2}{\mu_k^2}\right)\|g_k\|^2. \tag{26}$$

Summing eq. (26) over $k \in K^+$ and applying Lemma 6 yield

$$\sum_{k\in K^+}\delta_k \geq \sum_{k\in K^+}\left(\frac{\overline{\alpha}}{\mu_k} - \frac{M\overline{\alpha}+L/2}{\mu_k^2}\right)\|g_k\|^2$$
$$\geq \frac{\overline{\alpha}}{\theta_{\max}}\left(\sqrt{\varsigma+\sum_{k\in K^+}\|g_k\|^2} - \sqrt{\varsigma}\right) - \frac{M\overline{\alpha}+L/2}{\theta_{\min}^2}\left(\ln\left(\varsigma+\sum_{k\in K^+}\|g_k\|^2\right) - \ln(\varsigma)\right),$$

which yields the desired result. □

4.6 Global Convergence Rate

Finally, we combine the results for $k \in K^0$ stated in Lemma 5 and $k \in K^+$ stated in Lemma 7 to derive an upper bound on the total sum of $\|g_k\|^2$. We define a constant F as follows:

$$F \coloneqq f(x_0) - f^* + \frac{2}{1-\varepsilon_f}\left(\overline{f}(x_0) - \overline{f}^* + \Delta_{\max}\right) + C_1\sqrt{\varsigma} - C_2\ln(\varsigma). \tag{27}$$

Then, we can obtain the following unified bound of the sum of $\|g_k\|^2$.

Lemma 8 *Under Assumptions 2 to 4, the sum of* $\|g_k\|^2$ *over* $k \in K^0, K^+$ *satisfies*

$$F \geq \frac{c\overline{\alpha}}{M}\sum_{k\in K^0}\|g_k\|^2 + C_1\sqrt{\varsigma+\sum_{k\in K^+}\|g_k\|^2} - C_2\ln\left(\varsigma+\sum_{k\in K^+}\|g_k\|^2\right),$$

Proof Since $f(x_{k_{\max}}) \geq f^*$ by Assumption 2 and $K = K^0 \cup K^+$, we have

$$f(x_0) - f^* \geq f(x_0) - f(x_{k_{\max}}) = \sum_{k\in K}\delta_k = \sum_{k\in K^0}\delta_k + \sum_{k\in K^+}\delta_k.$$

From Lemmas 5 and 7, we can further bound as

$$f(x_0) - f^* \geq \frac{c\overline{\alpha}}{M} \sum_{k \in K^0} \|g_k\|^2 - \frac{2}{1-\varepsilon_{\mathrm{f}}} \left(\overline{f}(x_0) - \overline{f}^* + \Delta_{\max} \right)$$
$$+ C_1 \sqrt{\varsigma + \sum_{k \in K^+} \|g_k\|^2} - C_1 \sqrt{\varsigma} - C_2 \ln \left(\varsigma + \sum_{k \in K^+} \|g_k\|^2 \right) + C_2 \ln(\varsigma).$$

Substituting eq. (27) yields the desired bound, concluding the proof. □

Now, we establish the following convergence rate for Algorithm 1.

Theorem 1 *Under Assumptions 1 to 4, the average of squared gradient norms generated by Algorithm 1 satisfies*

$$\frac{1}{k_{\max}} \sum_{k=0}^{k_{\max}-1} \|g_k\|^2 = \mathscr{O}\left(\frac{1}{k_{\max}} \right).$$

Proof Let us define the function $\phi : \mathbb{R}_{>0} \to \mathbb{R}$ as

$$\phi(G) := \frac{C_1}{2} \sqrt{G} - C_2 \ln(G). \tag{28}$$

Since $\phi'(G) = \frac{C_1}{4\sqrt{G}} - \frac{C_2}{G}$, its minimum over $G > 0$ is attained at $G^* := \left(\frac{4C_2}{C_1} \right)^2$ with

$$\phi(G^*) = 2C_2 \left(1 - \ln \left(\frac{4C_2}{C_1} \right) \right) = 2C_2 \ln \left(\frac{\mathrm{e}C_1}{4C_2} \right),$$

where e is the base of the natural logarithm. Thus, Lemma 8 yields

$$F \geq \frac{c\overline{\alpha}}{M} \sum_{k \in K^0} \|g_k\|^2 + \frac{C_1}{2} \sqrt{\sum_{k \in K^+} \|g_k\|^2} + \phi \left(\sum_{k \in K^+} \|g_k\|^2 + \varsigma \right)$$
$$\geq \frac{c\overline{\alpha}}{M} \sum_{k \in K^0} \|g_k\|^2 + \frac{C_1}{2} \sqrt{\sum_{k \in K^+} \|g_k\|^2} + \phi(G^*),$$

which is equivalent to

$$0 \leq \frac{c\overline{\alpha}}{M} \sum_{k \in K^0} \|g_k\|^2 + \frac{C_1}{2} \sqrt{\sum_{k \in K^+} \|g_k\|^2} \leq F' \tag{29}$$

where

$$F' := F - \phi(G^*)$$
$$= f(x_0) - f^* + \frac{2}{1-\varepsilon_{\mathrm{f}}} \left(\overline{f}(x_0) - \overline{f}^* + \Delta_{\max} \right) + C_1 \sqrt{\varsigma} - C_2 \ln(\varsigma) - 2C_2 \ln \left(\frac{\mathrm{e}C_1}{4C_2} \right).$$

It remains to bound $\sum_{k \in K} \|g_k\|^2 = \sum_{k \in K^0} \|g_k\|^2 + \sum_{k \in K^+} \|g_k\|^2$ given the constraint in eq. (29). More generally, let $a, b > 0$, $c \geq 0$ and $x, y \geq 0$, and consider the maximization problem $x + y$ subject to $ax + b\sqrt{y} \leq c$. The maximum is achieved on the boundary $ax + b\sqrt{y} = c$ since $x + y$ is increasing in both variables. On this boundary, we can write $y = \left(\frac{c - ax}{b} \right)^2$, and thus $x + y$ is convex on an interval $0 \leq x \leq c/a$,

attaining its maximum at one of the endpoints. Evaluating at the endpoints yields the candidates $(x,y) = (c/a, 0)$ and $(x,y) = (0, (c/b)^2)$, and therefore the maximum of $x+y$ is $\max\left(c/a, (c/b)^2\right)$. Applying this result to eq. (29) gives

$$\begin{aligned}\frac{1}{k_{\max}} \sum_{k=0}^{k_{\max}-1} \|g_k\|^2 &= \frac{1}{k_{\max}} \left(\sum_{k \in K^0} \|g_k\|^2 + \sum_{k \in K^+} \|g_k\|^2 \right) \qquad (K = K^0 \cup K^+) \\ &\leq \frac{1}{k_{\max}} \max\left(\frac{MF'}{c\overline{\alpha}}, \frac{4F'^2}{C_1^2} \right), \end{aligned} \tag{30}$$

which completes the proof. □

As a remark, the function $\phi(G)$ in eq. (28) is closely related to the Lambert W function [9, 18]. For $x \in [-1/\mathrm{e}, 0)$, the equation $y\mathrm{e}^y = x$ admits two real negative solutions, and the smaller one is given by the second branch of Lambert W function $y = W_{-1}(x) < 0$. Using this function, the equation $\phi(G) = 0$ can be solved in closed form, which leads to a sharper bound in Theorem 1. Since this is not essential for the main argument, we omit the details and refer to [18].

We next interpret eq. (30) in the proof of Theorem 1. The contribution of the indices in K^0 is analogous to the classical analysis of gradient descent. Indeed, for gradient descent with fixed step size $\alpha = 1/L$, one has

$$\frac{1}{k_{\max}} \sum_{k=0}^{k_{\max}-1} \|g_k\|^2 \leq \frac{1}{k_{\max}} \frac{2(f(x_0) - f^*)}{\alpha}.$$

Thus, the term associated with K^0 reflects the standard $\mathcal{O}(f(x_0) - f^*)$ dependence of first-order methods since F' contains the initial gaps $f(x_0) - f^*$ and $\overline{f}(x_0) - \overline{f}^*$. In contrast, the contribution of K^+ resembles the behavior of AdaGrad-Norm, whose convergence bound scales quadratically with the initial optimality gap [44].

Finally, Theorem 1 implies that the iteration complexity:

$$\min\left\{k \;\middle|\; \|\nabla f(x_k)\| \leq \varepsilon_{\mathrm{gtol}}\right\} = \mathcal{O}\left(1/\varepsilon_{\mathrm{gtol}}^2\right).$$

This is a standard convergence guarantee for first-order optimization algorithms applied to L-smooth nonconvex functions. More explicitly, eq. (30) gives the sufficient bound

$$k_{\max} \geq \frac{1}{\varepsilon_{\mathrm{gtol}}^2} \max\left\{ \frac{MF'}{c\overline{\alpha}}, \frac{4(F')^2}{C_1^2} \right\}.$$

Here $\overline{\alpha}$ depends only on c, L, m, M and the backtracking bounds, $C_1 = \overline{\alpha}/\theta_{\max}$, $C_2 = (M\overline{\alpha} + L/2)/\theta_{\min}^2$, and F' is displayed in the proof and contains ε_{f}, ς, the initial true and observed optimality gaps, and $\Delta_{\max}$. Thus the order symbol suppresses no dependence on the iteration counter, but it does suppress these problem and algorithm parameters; the guarantee deteriorates as the permitted function error approaches 1 through the factor $(1 - \varepsilon_{\mathrm{f}})^{-1}$.

5 Practical Choices of Algorithmic Variables

In sections 3 and 4, we introduced Algorithm 1 and established its convergence properties. In this section, we discuss practical choices of algorithmic variables and parameters that lead to fast and stable performance. The quantities to be discussed here are the matrices B_k, the step sizes α_k, and the regularization parameters μ_k.

5.1 Approximate Hessian B_k

We first discuss the construction of the matrices B_k used in line 9 of Algorithm 1. The purpose of this subsection is to show that B_k can satisfy Assumption 4, the uniform spectral bound, with a slight modification and selection of curvature pairs using the L-BFGS method.

Construction of B_k: We briefly note the L-BFGS method and the curvature pair notation [4, 26, 31]. For an iteration k, we define

$$s_k \coloneqq x_{k+1} - x_k, \qquad y_k \coloneqq g_{k+1} - g_k$$

where g_k and g_{k+1} are the gradients at x_k and x_{k+1}, respectively. Starting from an initial matrix, an approximate Hessian is obtained by successively applying BFGS updates to a sequence of curvature pairs. Let $\{(s_{k_i}, y_{k_i})\}_{i=0}^{p-1}$ be a collection of curvature pairs, where k_i denotes the iteration index at which the i-th pair was generated. Starting from the initial matrix $B^{(0)} = \gamma I$ with $\gamma = \|y_{k_0}\|^2 / y_{k_0}^\top s_{k_0}$, we define $\mathrm{BFGS}(\{(s_{k_i}, y_{k_i})\}_{i=0}^{p-1})$ as the matrix $B^{(p)}$ obtained by sequentially applying the BFGS update as

$$B^{(j+1)} = B^{(j)} - \frac{B^{(j)} s_{k_j} s_{k_j}^\top B^{(j)}}{s_{k_j}^\top B^{(j)} s_{k_j}} + \frac{y_{k_j} y_{k_j}^\top}{y_{k_j}^\top s_{k_j}}$$

for $j = 0, 1, \ldots, p-1$, where each curvature pair (s_{k_j}, y_{k_j}) is applied in order. Here, $p > 0$ denotes the maximum number of stored curvature pairs and is a fixed small integer in the L-BFGS method. When fewer than p curvature pairs are available, the BFGS update is applied using all currently available pairs.

Now, we provide a sufficient condition for Assumption 4 by the following proposition, which is a variant of existing work [14, Theorem 5.5] [16, Lemma 1]. Its proof is deferred to appendix B.

Proposition 2 *Let $\{(s_{k_i}, y_{k_i})\}_{i=0}^{p-1}$ be a fixed number of curvature pairs with $s_{k_i} \neq 0$. Assume that there exist constants $0 < \lambda \leq \Lambda$ such that for all $(s, y) \in \{(s_{k_i}, y_{k_i})\}_{i=0}^{p-1}$,*

$$y^\top s \geq \frac{1}{\Lambda} \|y\|^2, \tag{31}$$

$$y^\top s \geq \lambda \|s\|^2. \tag{32}$$

Define the condition number $\kappa \coloneqq \Lambda / \lambda$. Then there exist $0 < m < M$ such that

$$mI \preceq \mathrm{BFGS}(\{s_{k_i}, y_{k_i}\}_{i=0}^{p-1}) \preceq MI, \tag{33}$$

where

$$M := (1+p)\Lambda, \qquad m := \left((1+\sqrt{\kappa})^{2p}\left(\frac{1}{\lambda}+\frac{1}{\lambda(2\sqrt{\kappa}+\kappa)}\right)\right)^{-1}.$$

Proposition 2 shows that, whenever eqs. (31) and (32) hold for the stored pairs, the resulting BFGS matrix satisfies Assumption 4 as expressed in eq. (33).

Based on Proposition 2, we form B_k from at most $p = 10$ stored modified curvature pairs. After damping, a pair $(s,\bar{y})$ is retained only if

$$\bar{y}^\top s \geq \max\left\{10^{-8}\|s\|^2, \frac{\|\bar{y}\|^2}{10^8}, 10^{-16}\right\}. \tag{34}$$

Thus every stored pair satisfies eqs. (31) and (32) with the iteration-independent constants $\lambda = 10^{-8}$ and $\Lambda = 10^8$. The finite memory bound and Proposition 2 then provide uniform constants m and M, independent of k, for the matrices actually used by the implementation. The absolute floor in eq. (34) rejects numerically degenerate pairs but is not needed for this implication.

We use

$$B_k = \mathrm{BFGS}(\{(s_{k_i},\bar{y}_{k_i})\}), \tag{35}$$

where

$$\bar{y}_{k_i} = \theta_i^{\mathrm{damp}} y_{k_i} + (1-\theta_i^{\mathrm{damp}}) B_{k-1} s_{k_i}, \qquad \theta_i^{\mathrm{damp}} \in [0,1].$$

We determine θ_i^{damp} using the standard damped BFGS construction and its limited-memory variant [27, 34] before applying eq. (34). This explicit two-sided test, rather than convexity or cocoercivity of ∇f, establishes consistency between the nonconvex theory and implementation.

Remarks for Practical Implementation: Having established that the proposed construction of B_k satisfies Assumption 4, we now present several remarks on its practical implementation. We begin by describing how to compute the search direction

$$d_k = -(B_k + \mu_k I)^{-1} g_k$$

with B_k in eq. (35). When $\mu_k = 0$, the direction can be computed efficiently by the standard L-BFGS two-loop recursion [26, 31]. For the case $\mu_k > 0$, regularized limited-memory schemes may in principle be employed [24]. Alternatively, one may adopt the simple and robust approximation mentioned in the same reference. Specifically, we approximate B_k in eq. (35) by

$$B_k \approx \mathrm{BFGS}(\{s_{k_i}, \bar{y}_{k_i} + \mu_k s_{k_i}\}_{i=0}^{p-1}) - \mu_k I, \tag{36}$$

so that $B_k + \mu_k I \approx \mathrm{BFGS}(\{s_{k_i}, \bar{y}_{k_i} + \mu_k s_{k_i}\}_{i=0}^{p-1})$. Its inverse can then be computed using the standard two-loop recursion. Although this approximation is inexact, it has been reported to yield nearly identical practical performance while remaining comparatively simple to implement [24]. We adopt this approximation in our numerical experiments.

Separately, there exist function-based modified secant conditions [1, 21, 46, 49] that adjust the secant equation using available function values when those quantities are reliable. Such adjustments aim to improve practical convergence behavior. We partially adopt these function-based modifications following prior work [47, 50]. The suffix `-MS` in section 6 indicates the use of this technique. Because these modifications are well established, we omit low-level implementation details here. Further specifics are available in the cited references and in our open-source implementation.

5.2 Step Size α_k

We next discuss the adjustment of the step size α_k in line 4 of Algorithm 2. As indicated in the pseudo-code, we choose α_k using interpolation of the objective function. Moré–Thuente line searches [28] are widely used in quasi-Newton methods [42], and they typically employ cubic or quadratic interpolation to select trial step sizes. In our implementation, we use an essentially identical procedure, except that the trial step size is obtained by clipping it to the interval $[\beta_{\min}\alpha_k, \beta_{\max}\alpha_k]$. For all numerical experiments, we set the Armijo parameter to $c = 10^{-4}$ and choose the interpolation bounds as $\beta_{\min} = 1/16$ and $\beta_{\max} = 15/16$.

Furthermore, when $\mu_k > 0$ and $\alpha_k = 1$, we introduce a one-time adjustment of the trial step size using gradient information. Let d_k be the search direction and g_k and g_{try} the gradients at the current and trial points. If the directional derivative along d_k changes sign, $d_k^\top g_k < 0$ and $d_k^\top g_{\text{try}} > 0$, and the trial gradient is sufficiently aligned with d_k, namely $d_k^\top g_{\text{try}} > 0.5\|d_k\|\|g_{\text{try}}\|$, we rescale α_k once by a secant-like factor,

$$\alpha_k \leftarrow \operatorname{clip}\left(\alpha_k \frac{-d_k^\top g_k}{d_k^\top g_{\text{try}} - d_k^\top g_k}, \beta_{\min}\alpha_k, \beta_{\max}\alpha_k\right),$$

where $\operatorname{clip}(x, a, b) := \min(\max(x, a), b)$ is a clipping function. When objective values are unreliable and backtracking and interpolation do not occur, this correction effectively refines the step size. Since this adjustment is at most once per iteration, the theoretical results in section 4 remain valid.

5.3 Regularization Parameter μ_k

We finally discuss the adjustment of the regularization parameter μ_k in line 8 of Algorithm 1. In our implementation, the OFFO accumulator contains exactly those gradient norms generated during regularized iterations:

$$G_k^2 = \varsigma + \sum_{j \in K^+, j \le k} \|g_j\|^2, \qquad \mu_k = \operatorname{clip}\left(\frac{1}{10}\|g_k\|, \frac{1}{100}G_k, G_k\right).$$

Hence $\mu_k = \theta_k G_k$ with $\theta_k \in [10^{-2}, 1]$, exactly as required by eq. (7). We set $\varsigma = 10^{-20}$, so that its square root, the initial scale G_{-1}, equals 10^{-10}. The primary implementation does not restart this accumulator; the earlier heuristic restart is disabled by default and is considered only as an ablation. This removes the unproved finite-restart qualification from the convergence claim.

6 Numerical Experiments

We conducted numerical experiments to evaluate the practical performance of our proposed method and to compare it with existing optimization algorithms.

6.1 Methods

We compared the following methods in our experiments.

- (`Ours`): Our proposed regularized quasi-Newton method
  - We used Algorithms 1 and 2 with the details in section 5.
  - (`Ours-MS`) means the variant incorporating the modified secant condition described in section 5.1.
- (`Line`): Line search L-BFGS method
  - Our Python implementation ported from LBFGSpp [32].
  - The step size was determined using the Moré–Thuente line search, closely resembling SciPy's implementation.
  - (`Line-MS`) means the same variant as `Ours-MS`.
- (`Reg`): Regularized L-BFGS method [24]
  - A quadratic regularization-based quasi-Newton method that does not rely on line search and is conceptually closer to trust-region methods.
  - We wrapped the function `regLBFGS` with `solveNonmonotone` from [24].
  - (`Reg-Sec`) means the incorporation of the approximated computation as in eq. (36). This is a wrapped function of `regLBFGSsec` from [24].
- (`SciPy`): SciPy's L-BFGS-B method [42]
  - An established L-BFGS method implemented in C and called from Python.
  - We used the default parameters, with the exception that `ftol` was set to 0 to avoid premature termination.
- (`NTQN`): Noise-tolerant quasi-Newton method [38, 45]
  - A method designed to accommodate inexact gradient information, as mentioned in section 2.3.
  - We report both the original wrapper setting and the authors' recommended termination mode (`terminate=3`).
  - For the latter, the external solver controls termination; the common gradient-tolerance criterion is used only when constructing performance profiles and is not injected into the solver.

In the following, each method is referred to by the name given in parentheses. All these methods are L-BFGS-based, and we set the number of stored curvature pairs to 10, the default value in SciPy's L-BFGS-B implementation. All experiments were conducted on a Microsoft Windows 11 Home system (version 10.0, build 26100) equipped with a 13th Gen Intel® Core™ i7-1360P CPU, featuring 12 physical cores and 16 logical processors. All computations were performed using the CPU only.

explanation	min abs value	relative error	ε_{f}
64-bit double precision	2.23×10^{-308}	$2^{-52} \approx 2.22 \times 10^{-16}$	2.22×10^{-9}
32-bit single precision	1.18×10^{-38}	$2^{-23} \approx 1.19 \times 10^{-7}$	1.19×10^{-3}
16-bit half precision	6.10×10^{-5}	$2^{-10} \approx 9.77 \times 10^{-4}$	9.77×10^{-2}

Table 1: The "min abs value" is the smallest positive normalized number, the "relative error" is the maximum relative rounding error, and ε_{f} is the parameter in (2) used in our experiments.

6.2 Experimental Results

6.2.1 Test Problems and Settings

We evaluated all algorithms on unconstrained problems from the CUTEst benchmark collection [15], implemented in Python using the PyCUTEst interface [13]. All problems were initialized with the default starting points provided by the library. Following the previous work [24], we excluded problems such as the initial point being already stationary or containing `NaN` and `INF` in its function values or gradient. The 64- and 32-bit collections contain 220 problems with dimensions from 2 to 123,200, and the 16-bit collection contains 152 problems over the same dimension range. The supplied error rates are 2.22×10^{-9}, 1.19×10^{-3}, and 9.77×10^{-2} for the 64-, 32-, and 16-bit experiments, respectively. In addition to these precision experiments, we distinguish two additive-noise experiments on the 64-bit collection:

1. a theory-aligned setting with $\overline{f}(x) = f(x) + \xi_f$, $\xi_f \sim \mathrm{unif}(-10^{-3}, 10^{-3})$, and exact gradients; and
2. an adverse stress test with the same function perturbation and independent componentwise gradient perturbations $\xi_g^{(i)} \sim \mathrm{unif}(-10^{-3}, 10^{-3})$.

We supply $\varepsilon_{\mathrm{f}} = 10^{-2}$ in both settings and additionally test 10^{-4} and 10^{-1} to assess under- and overestimation of this error parameter. Each noisy experiment records the random seed, and all solvers receive the same seed for a given problem and scenario. The joint-noise setting deliberately violates eq. (3); it is included to test performance under an unfavorable perturbation, not to validate the exact-gradient theorem.

Each setting uses a different error rate ε_{f} in the error model (eq. (2)), as summarized in Table 1. The table relates the machine epsilon (maximum relative rounding error) for each floating-point precision to the ε_{f} value used in our experiments. The relatively large ε_{f} values are chosen to account for the cumulative errors that may arise during optimization. We also set the maximum number of iterations to $k_{\max} = 15{,}000$ and timed out runs exceeding 10 minutes per problem.

6.2.2 Performance Profiles

We used performance profiles [11] to compare the algorithms. Let P denote the set of test problems and S the set of solvers. For each problem $p \in P$ and solver $s \in S$, let $n_{p,s} > 0$ denote the number of oracle calls (i.e., function and gradient evaluations)

required to reach a specified gradient norm tolerance $\varepsilon_{\text{gtol}}$. In this experiment, we use the number of oracle calls rather than clock time to avoid the influence of algorithmic overhead and to focus on the intrinsic optimization efficiency. When a solver fails to converge within the gradient tolerance or exceeds a maximum number of oracle calls, we set $n_{p,s} = +\infty$. Then, the performance ratio $r_{p,s}$ is defined as

$$r_{p,s} = \frac{n_{p,s}}{\min_{s' \in S} n_{p,s'}}.$$

For a given threshold $\tau \geq 1$ and solver s, the performance profile plots the proportion of problems for which $r_{p,s} \leq \tau$. This means that the y-coordinate for solver s at x-coordinate τ is given by

$$\rho_s(\tau) = \frac{1}{|P|} \left| \{ p \in P \mid r_{p,s} \leq \tau \} \right|.$$

A curve that lies above others indicates better performance. Following standard implementations [42], we used the infinity norm for the gradient, though other norms may equally be employed.

6.2.3 Results with Artificial Noise

We first separate the theory-aligned function-noise experiment from the joint-noise stress test described in section 6.2.1. The former tests the assumptions of section 4; the latter asks whether the implementation remains useful when both available quantities are perturbed. Therefore, success in the joint-noise experiment is empirical evidence of robustness beyond the theorem and is not presented as validation of the exact-gradient complexity result. The experiment notebook also compares the hybrid method with an always-OFFO variant and the earlier accumulator-restart heuristic, and varies the supplied ε_{f} over two orders of magnitude around its nominal value.

6.2.4 Results at Different Precision Levels

We next present performance profiles for each floating-point precision (64-, 32-, and 16-bit). We evaluated each method at three gradient norm tolerance levels, i.e., $\varepsilon_{\text{gtol}} \in \{10^{-1}, 10^{-3}, 10^{-5}\}$. These tolerance levels represent different optimization regimes, from moderate accuracy to high precision requirements. Figure 2 shows the comprehensive performance profiles across all three precision levels. The profiles show that `Ours` and `Ours-MS` solve a large fraction of the retained problems across the three precision levels. Their relative ranking depends on the tolerance and precision, so we avoid claiming uniform superiority and report both the fraction solved and the performance ratios in Figure 2.

6.3 Computational Time per Iteration

In this subsection, we present an experiment to compare the per-iteration computational cost of each method. Here, our purpose lies in the measurement of the total

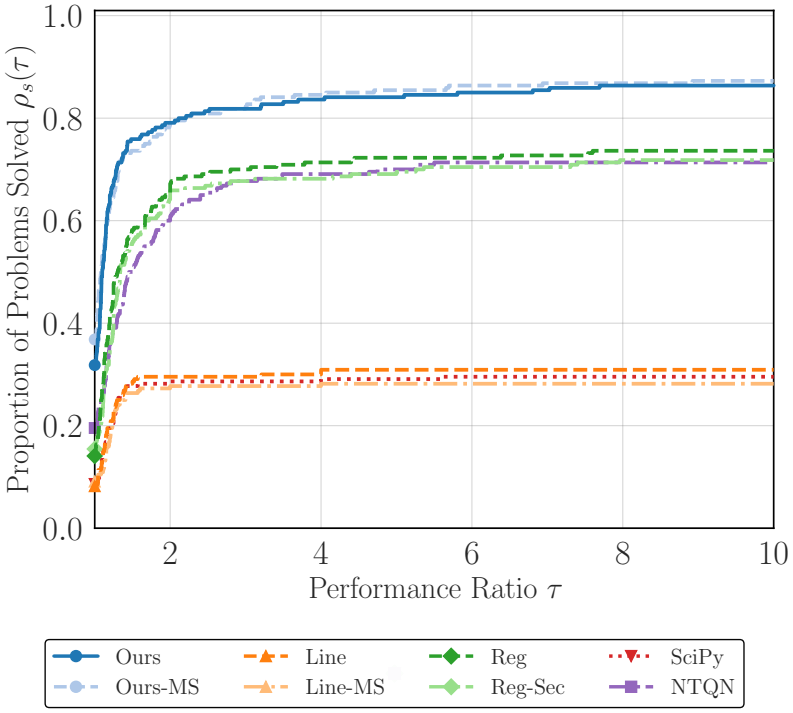


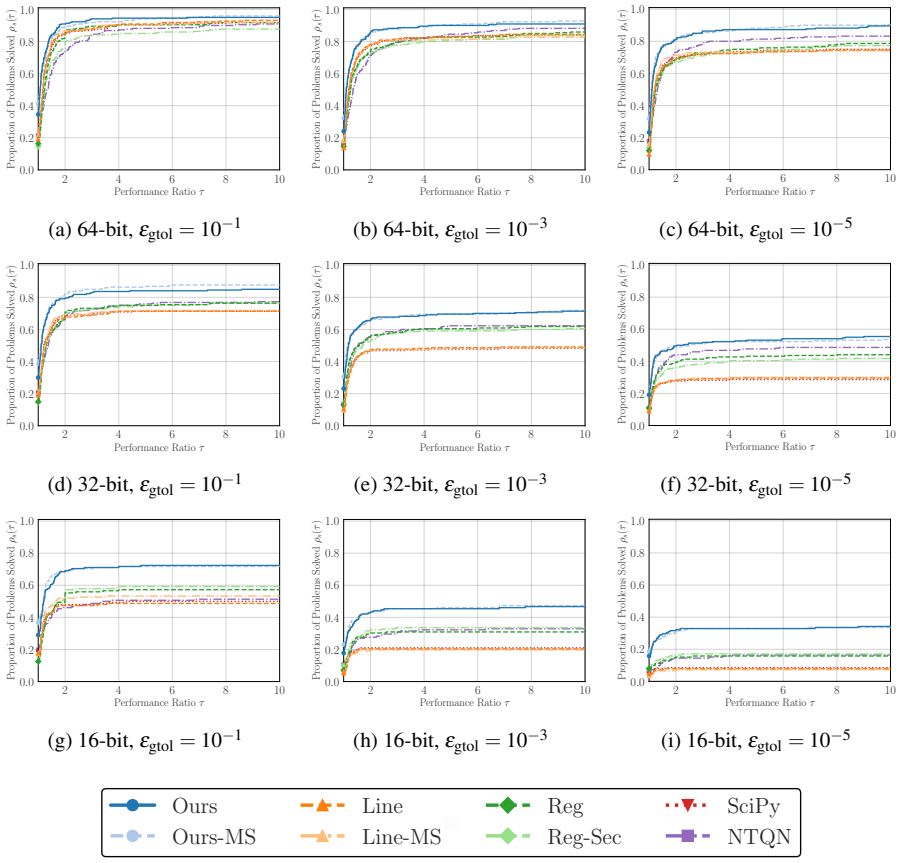
Fig. 1: Performance profile for the deliberately adverse joint function-and-gradient noise setting (componentwise level 10^{-3}) and $\varepsilon_{\text{gtol}} = 10^{-2}$. This experiment lies outside the exact-gradient theory.

execution time rather than the number of oracle calls. We conducted experiments on the ill-conditioned quadratic problem

$$\underset{x\in\mathbb{R}^n}{\text{minimize}} \quad \frac{1}{2}\sum_{i=1}^{n} i x_i^2$$

with $n = 10{,}000$. The initial point x_0 was chosen as the all-ones vector. We used memory size $m = 10$ and maximum iterations $k_{\max} = 100$, and we did not set any stopping criteria other than reaching $k_{\max}$. The simplicity of the optimization problem and the small value of $k_{\max}$ are well suited to the purpose of this experiment, as they ensure that the total runtime is dominated by algorithmic overhead rather than function or gradient computation. Timing data were collected after three warm-up runs, followed by 100 recorded trials.

In Figure 3, we report the execution time statistics, and we can observe that our proposed methods (`Ours`) and (`Ours-MS`) exhibit competitive performance compared to other methods. Combining with the results from section 6.2, our methods are competitive or superior in total oracle calls and per-iteration computational cost, indicating practical efficiency. As a side note, although the (`SciPy`) calls optimized C



(a) 64-bit, $\varepsilon_{\text{gtol}} = 10^{-1}$ (b) 64-bit, $\varepsilon_{\text{gtol}} = 10^{-3}$ (c) 64-bit, $\varepsilon_{\text{gtol}} = 10^{-5}$

(d) 32-bit, $\varepsilon_{\text{gtol}} = 10^{-1}$ (e) 32-bit, $\varepsilon_{\text{gtol}} = 10^{-3}$ (f) 32-bit, $\varepsilon_{\text{gtol}} = 10^{-5}$

(g) 16-bit, $\varepsilon_{\text{gtol}} = 10^{-1}$ (h) 16-bit, $\varepsilon_{\text{gtol}} = 10^{-3}$ (i) 16-bit, $\varepsilon_{\text{gtol}} = 10^{-5}$

Fig. 2: Performance profiles across different floating-point precisions (64-, 32-, and 16-bit) and tolerance levels. The results demonstrate the robustness of the proposed methods.

routines, it incurs associated dispatch overhead, producing a larger mean time. One of the reasons why the (`Reg`) and (`Reg-Sec`) methods take longer execution time is due to the implementation aspect of the original code.

In Table 2, we report the total number of oracle calls (calls of $\overline{f}$ and $\nabla\overline{f}$) and final objective values $\overline{f}(x_{k_{\max}})$. The purpose of this table is to confirm that all methods achieved similar convergence behavior within the limited number of iterations. The comparable values in the table indicate that the comparison of execution times is fair and not affected by differences in convergence behavior.

7 Conclusion

Fig. 3: Mean execution times for the experiment in section 6.3. This plot roughly indicates the extra computational overhead of each method. Error bars represent one standard deviation over 100 runs.

	Line	Ours	Line-MS	Ours-MS	NTQN	SciPy	Reg-Sec	Reg
# Calls	212	202	212	202	219	212	206	206
$\overline{f}(x_{k_{\max}})$	1.24	1.34	1.24	1.34	1.43	1.24	1.48	1.47

Table 2: The number of oracle calls and final observed objective values for the experiment in section 6.3.

We proposed a hybrid regularized quasi-Newton method for smooth nonconvex optimization with bounded errors in objective-function values. Under exact gradients and explicit uniform spectral bounds on the quasi-Newton matrices, the method has the standard $\mathcal{O}(1/\varepsilon^2)$ first-order iteration complexity. The implementation enforces the required two-sided curvature conditions directly and uses the same non-restarted OFFO accumulator as the analysis. The experiments distinguish function-value noise covered by the theorem from a joint-noise stress test outside its scope and assess sensitivity to the supplied error bound. An analysis with inexact gradients, which can at best guarantee convergence to a noise-dependent neighborhood without additional assumptions, remains future work.

A Proof of Lemma 6

The proof of Lemma 6 is similar to existing results [44, Lemma 3.2], [18, Lemma 3.1].

Proof Recall that eq. (7) defines μ_k as $\theta_k\sqrt{\varsigma+\sum_{j\in K^+,\, j\le k}\left\|g_j\right\|^2}$, and line 8 of Algorithm 1 asserts $\theta_k \in [\theta_{\min}, \theta_{\max}]$. Let $k_i \in K^+$ be the i-th element of K^+. For any

$0 < x < y$, $x/y \le -\ln(1-x/y) = \ln y - \ln(y-x)$ holds. Thus, we have

$$\begin{aligned}\frac{\left\|g_{k_i}\right\|^2}{\mu_{k_i}^2} &\le \frac{1}{\theta_{\min}^2}\frac{\left\|g_{k_i}\right\|^2}{\varsigma+\sum_{\ell=0}^{i}\left\|g_{k_\ell}\right\|^2} && \text{(definition)}\\ &\le \frac{1}{\theta_{\min}^2}\left(\ln\left(\varsigma+\sum_{\ell=0}^{i}\left\|g_{k_\ell}\right\|^2\right)-\ln\left(\varsigma+\sum_{\ell=0}^{i-1}\left\|g_{k_\ell}\right\|^2\right)\right). && (x/y \le \ln y-\ln(y-x))\end{aligned}$$

Similarly, $x/\sqrt{y} \ge x/(\sqrt{y}+\sqrt{y-x}) = \sqrt{y}-\sqrt{y-x}$ holds. Thus, we have

$$\begin{aligned}\frac{\left\|g_{k_i}\right\|^2}{\mu_{k_i}} &\ge \frac{1}{\theta_{\max}}\frac{\left\|g_{k_i}\right\|^2}{\sqrt{\varsigma+\sum_{\ell=0}^{i}\left\|g_{k_\ell}\right\|^2}} && \text{(definition)}\\ &\ge \frac{1}{\theta_{\max}}\left(\sqrt{\varsigma+\sum_{\ell=0}^{i}\left\|g_{k_\ell}\right\|^2}-\sqrt{\varsigma+\sum_{\ell=0}^{i-1}\left\|g_{k_\ell}\right\|^2}\right). && (x/\sqrt{y} \ge \sqrt{y}-\sqrt{y-x})\end{aligned}$$

Summing these inequalities over $0 \le i < |K^+|$ yields the claim. □

B Proof of Proposition 2

The proof of Proposition 2 is similar to existing results [14, Theorem 5.5] [16, Lemma 1].

Proof We consider standard BFGS updates with a single curvature pair (s,y). From eqs. (31) and (32), we have

$$\left\|\frac{yy^\top}{y^\top s}\right\| = \frac{y^\top y}{y^\top s} \le \Lambda, \quad \left\|\frac{ys^\top}{y^\top s}\right\| \le \frac{\|y\|\|s\|}{\sqrt{\frac{1}{\Lambda}\|y\|^2}\sqrt{\lambda\|s\|^2}} = \sqrt{\kappa}, \quad \left\|\frac{ss^\top}{y^\top s}\right\| = \frac{s^\top s}{y^\top s} \le \frac{1}{\lambda}. \tag{37}$$

We next bound the scalar γ used in the initial matrix γI. From eq. (31), we have $\gamma \le \Lambda$. By Cauchy–Schwarz inequality and eq. (32), we have $\gamma \ge \left\|y_{k_0}\right\|^2/\left\|y_{k_0}\right\|\left\|s_{k_0}\right\| = \left\|y_{k_0}\right\|/\left\|s_{k_0}\right\| \ge \lambda$. Thus, γ is bounded by $\lambda \le \gamma \le \Lambda$.

Let B, H denote the positive definite Hessian approximation and its inverse before the update, and B_+, H_+ the updated matrices. The BFGS update rule is

$$B_+ = B - \frac{Bss^\top B}{s^\top Bs} + \frac{yy^\top}{y^\top s}, \quad H_+ = \left(I - \frac{sy^\top}{y^\top s}\right)H\left(I - \frac{ys^\top}{y^\top s}\right) + \frac{ss^\top}{y^\top s}.$$

Since B is symmetric positive definite matrix and $s \ne 0$, we have

$$\begin{aligned}\|B_+\| &\le \|B\| + \left\|\frac{yy^\top}{y^\top s}\right\| \le \|B\| + \Lambda, \qquad (38)\\ \|H_+\| &= \left\|\left(I - \frac{ys^\top}{y^\top s}\right)^\top H\left(I - \frac{ys^\top}{y^\top s}\right) + \frac{ss^\top}{y^\top s}\right\|\end{aligned}$$

$$
\begin{aligned}
&\leq \left\| I - \frac{ys^\top}{y^\top s} \right\|^2 \|H\| + \left\| \frac{ss^\top}{y^\top s} \right\| \\
&\leq \left(1 + \left\| \frac{ys^\top}{y^\top s} \right\| \right)^2 \|H\| + \left\| \frac{ss^\top}{y^\top s} \right\| \\
&\leq \left(1+\sqrt{\kappa}\right)^2 \|H\| + \frac{1}{\lambda}. \qquad \text{(eq. (37))} \qquad (39)
\end{aligned}
$$

Recall that $\mathrm{BFGS}(\{(s_{k_i}, y_{k_i})\}_{i=0}^{p-1})$ denotes the matrix obtained by sequentially applying the BFGS update to the initial matrix γI using the curvature pairs $\{(s_{k_i}, y_{k_i})\}_{i=0}^{p-1}$. Applying eqs. (38) and (39) recursively over p updates and using the initial matrix γI, we obtain

$$
\begin{aligned}
\left\| \mathrm{BFGS}(\{s_{k_i}, y_{k_i}\}_{i=0}^{p-1}) \right\| &\leq \gamma + p\Lambda \leq (1+p)\Lambda, \\
\left\| (\mathrm{BFGS}(\{s_{k_i}, y_{k_i}\}_{i=0}^{p-1}))^{-1} \right\| &\leq (1+\sqrt{\kappa})^{2p}\gamma^{-1} + \frac{1}{\lambda}\sum_{i=0}^{p-1}(1+\sqrt{\kappa})^{2i} \\
&\leq (1+\sqrt{\kappa})^{2p}\frac{1}{\lambda} + \frac{1}{\lambda}\frac{(1+\sqrt{\kappa})^{2p}-1}{(1+\sqrt{\kappa})^2-1} \\
&\leq (1+\sqrt{\kappa})^{2p}\left(\frac{1}{\lambda} + \frac{1}{\lambda\left(2\sqrt{\kappa}+\kappa\right)}\right).
\end{aligned}
$$

Therefore, $\mathrm{BFGS}(\{s_{k_i}, y_{k_i}\}_{i=0}^{p-1})$ satisfies the stated bounds in eq. (33). □

The two curvature inequalities are standard sufficient conditions for bounded limited-memory BFGS matrices [5, Theorem 2.1]. In the implementation they are enforced directly by eq. (34); no convexity or cocoercivity assumption is used. This point is important because the convergence theorem permits nonconvex objectives.

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Statements and Declarations

Funding

This work was partially supported by JSPS KAKENHI (23H03351, 24K23853) and JST CREST (JPMJCR24Q2).

Competing Interests

The authors have no relevant financial or non-financial interests to disclose.

Author Contributions

All authors contributed to the study conception and design. The first draft of the manuscript was written by Hiroki Hamaguchi and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Data Availability

All data analyzed during this study are publicly available. URLs are included in this article.

Code Availability

The code used in this study is publicly available at the following GitHub repository: https://github.com/HirokiHamaguchi/qnlab.